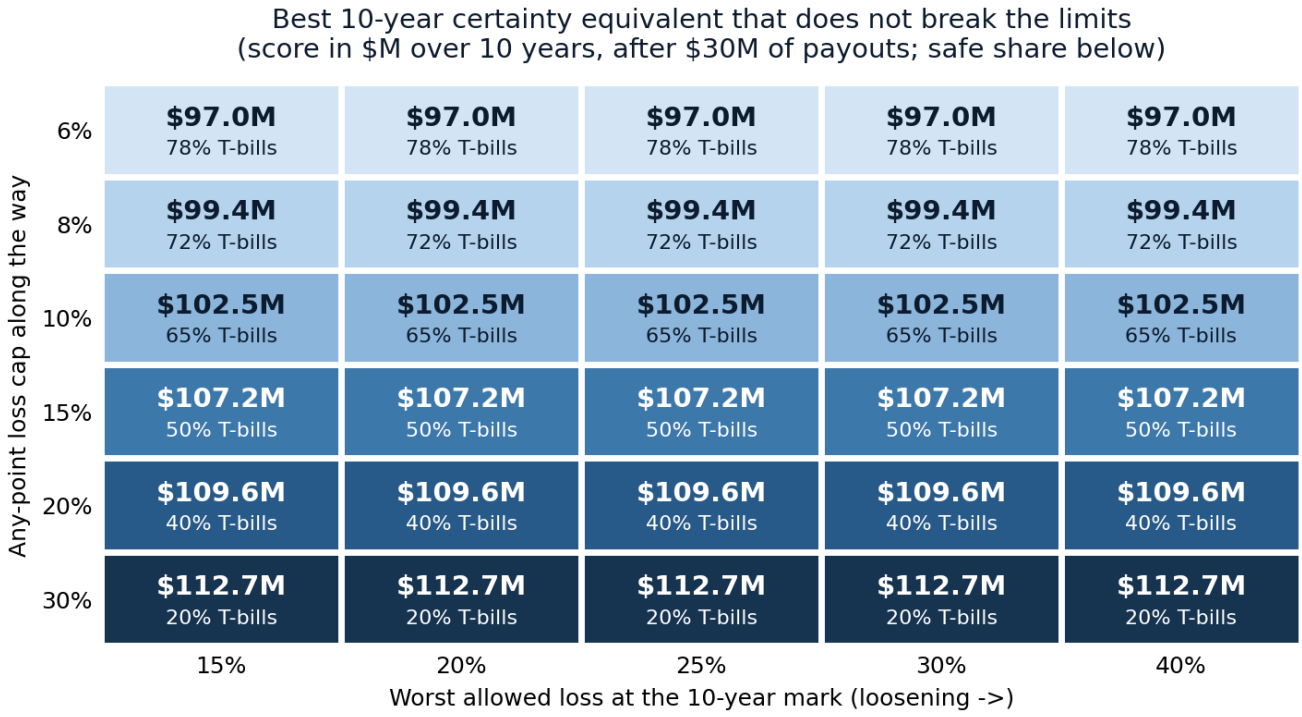
How much do the two loss limits cost over 10 years?

You asked for a matrix of the two loss budgets at a 10-year horizon, loosening both from their current strictest values, with the best portfolio that stays inside each pair of limits. Labeled sensitivity: only the 1-year, $250k/month case is confirmed; 10 years x $250k/month ($30M total) is exploratory.

The headline in one line

At 10 years, only the any-point drawdown cap matters. The year-10 loss cap never binds anywhere in the 15-40% band: after a decade of growth, even a 90%-risky mix ends below the starting $100M in at most an 8% tail bound, far inside every column. So every row of the matrix is one number, and the column choice is free within this range. Loosening the drawdown cap from 6% to 30% lifts the 10-year certainty equivalent from $97.0M to $112.7M (+$15.7M) and cuts T-bills from 78% to 20% of the portfolio.



Each cell is the best-scoring mix on fresh validation paths, then re-checked on independent final paths. Score = CARA certainty equivalent of 10-year leftover wealth plus payout cover, same ledger convention as the unified report; it is a 10-year score, not comparable to the one-year CE figures. Every cell passes the strict all-payments check (funded on every final path, zero missed payments).

What each loosening buys you

Any-point cap	T-bills	VTI	Gold	IEF	Property	Expected return	Score (CE)	Median left	5% bad case
6% (current)	78%	3.3%	2.9%	6.3%	10%	3.0%/yr	$97.0M	$101M	$86M
8%	72%	4.6%	4.1%	8.8%	10%	3.3%/yr	$99.4M	$105M	$87M
10%	65%	6.6%	5.9%	12.6%	10%	3.7%/yr	$102.5M	$112M	$88M
15%	50%	10.5%	9.4%	20.1%	10%	4.6%/yr	$107.2M	$123M	$92M
20%	40%	13.1%	11.8%	25.1%	10%	5.2%/yr	$109.6M	$131M	$93M
30%	20%	18.3%	16.5%	35.2%	10%	6.5%/yr	$112.7M	$147M	$97M

Rows are identical across the 15-40% year-10 loss columns, so one row per any-point cap. Median left / 5% bad case: leftover after all $30M of payouts, on the independent final paths. Expected return: model-fitted annual portfolio return. All rows: every payment funded on every final path; independent final check passed with zero walk-back.

How to read this

- The drawdown cap is your real risk dial. It alone sets the safe share: 6% forces 78% T-bills; 30% allows 20%. The year-10 loss cap only starts to bind below roughly 8% (the tail bound of the most aggressive mix), outside the range tested.
- Payments are safe everywhere in this matrix: $250k/month is about 3% of wealth per year, and every mix funds every payment on every simulated path. This matrix is purely about how much short-term pain you accept along the way.
- The trade is concrete: accepting a possible 30% dip instead of 6% adds about $15.7M of 10-year score, raises the median leftover from $101M to $147M, and even the 5% bad case improves from $86M to $97M. The cost is living through the deeper dips.

Method and limits

- Mix search, not full optimization: the engine optimizer is priced for one year, so each cell searches safe shares from 10% to 90% in 2.5% steps. The risky sleeve keeps the one-year optimizer proportions (VTI : gold : Treasuries = 26 : 24 : 50, no EM) and property stays fixed at 10%. A true 10-year optimizer might find slightly better mixes inside each cell.
- Frozen validation discipline: selection on 4 x 200 fresh validation paths, then an independent final check on 4 x 400 disjoint paths. All 30 cells passed the final check with no step-back. The binding check at every frontier is the any-point path-loss tail, exactly as in the 1-year case.
- Same model caveats as the 4th edition, amplified by horizon: weekly iid mixture law fitted to history (volatility clustering grows more relevant over 10 years), nominal dollars (no inflation indexation), T-bill yield fitted at its 2005-2026 average of 1.8% (today's ~4% would lift every cell), and a fixed 10% illiquid property sleeve. Monte Carlo diagnostics, not economic guarantees.

Source: studies/tbill/study_matrix.py, results/study_matrix.json on branch tbill-asset (private fork). Draws shared across all cells (same seeds), so differences between cells are purely the limits, not simulation noise.